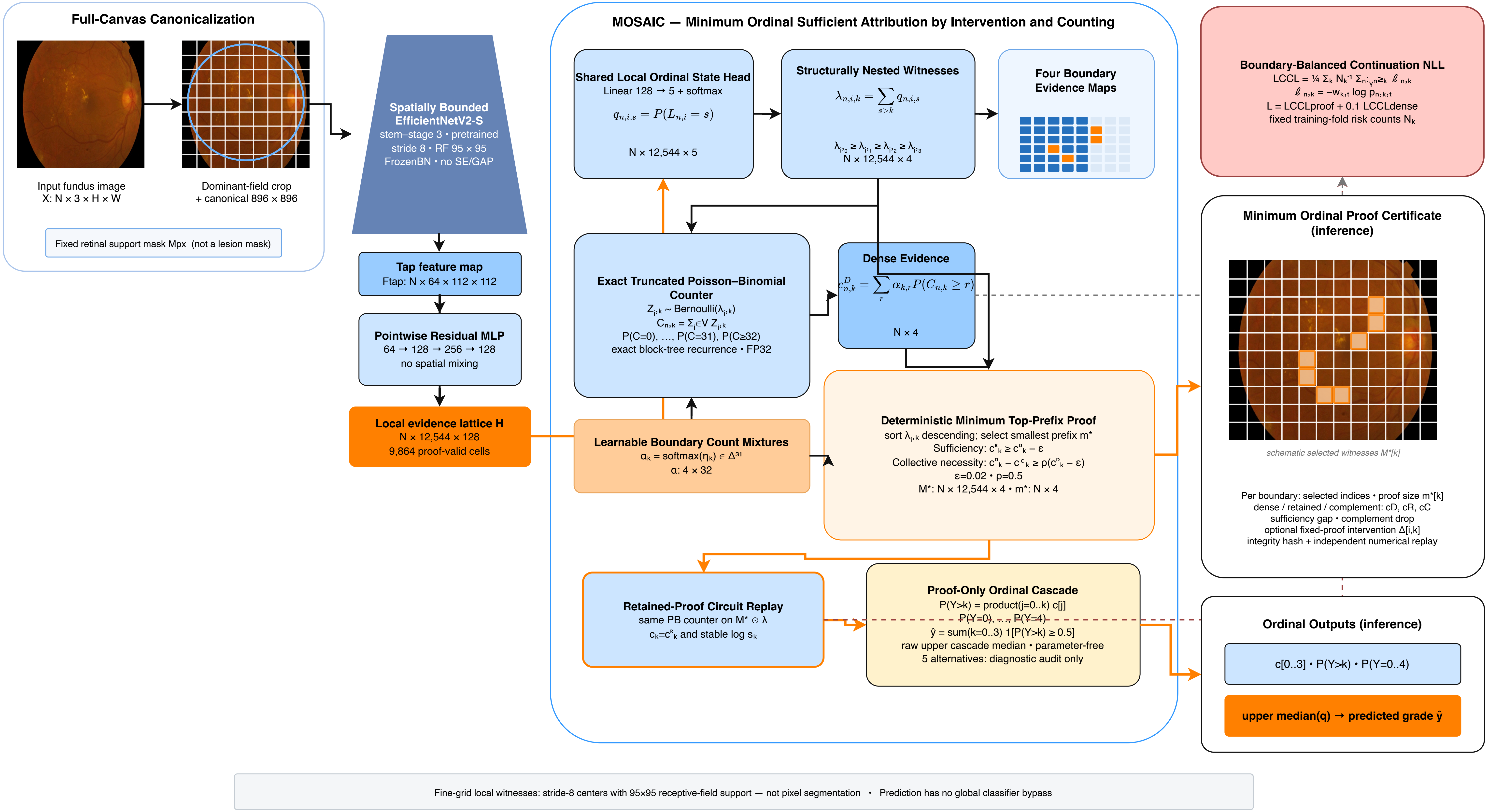
Full-Canvas Canonicalization
Input fundus image
X: N × 3 × H × W
Dominant-field crop
+ canonical 896 × 896
Fixed retinal support mask Mpx (not a lesion mask)
Spatially Bounded EfficientNetV2-S
stem–stage 3 • pretrained
stride 8 • RF 95 × 95
FrozenBN • no SE/GAP
Tap feature map
Ftap: N × 64 × 112 × 112
Pointwise Residual MLP
64 → 128 → 256 → 128
no spatial mixing
Local evidence lattice H
N × 12,544 × 128
9,864 proof-valid cells
MOSAIC — Minimum Ordinal Sufficient Attribution by Intervention and Counting
Shared Local Ordinal State Head
Linear 128 → 5 + softmax
$q_{n,i,s} = P(L_{n,i} = s)$
N × 12,544 × 5
Structurally Nested Witnesses
$\lambda_{n,i,k} = \sum_{s>k} q_{n,i,s}$
$\lambda_{i,0} \geq \lambda_{i,1} \geq \lambda_{i,2} \geq \lambda_{i,3}$
N × 12,544 × 4
Four Boundary Evidence Maps
Exact Truncated Poisson–Binomial Counter
$Z_{i,k} \sim \mathrm{Bernoulli}(\lambda_{i,k})$
$C_{n,k} = \Sigma_{i \in V} Z_{i,k}$
P(C=0), …, P(C=31), P(C≥32)
exact block-tree recurrence • FP32
Dense Evidence
$c^D_{n,k} = \sum_r \alpha_{k,r} P(C_{n,k} \geq r)$
N × 4
Learnable Boundary Count Mixtures
$\alpha_k = \mathrm{softmax}(\eta_k) \in \Delta^{31}$
α: 4 × 32
Deterministic Minimum Top-Prefix Proof
sort $\lambda_{i,k}$ descending; select smallest prefix m*
Sufficiency: $c^R_k \geq c^D_k - \varepsilon$
Collective necessity: $c^D_k - c^C_k \geq \rho(c^D_k - \varepsilon)$
ε=0.02 • ρ=0.5
M*: N × 12,544 × 4 • m*: N × 4
Retained-Proof Circuit Replay
same PB counter on M* ⊙ λ
$c_k = c^R_k$ and stable log s_k
Proof-Only Ordinal Cascade
P(Y>k) = product(j=0..k) c[j]
P(Y=0), …, P(Y=4)
ŷ = sum(k=0..3) 1[P(Y>k) ≥ 0.5]
raw upper cascade median • parameter-free
5 alternatives: diagnostic audit only
Boundary-Balanced Continuation NLL
$\mathrm{LCCL} = \frac{1}{4} \Sigma_k N_k^{-1} \Sigma_{n:y_n \geq k} \ell_{n,k}$
$\ell_{n,k} = -w_{k,t} \log p_{n,k,t}$
L = LCCLproof + 0.1 LCCLdense
fixed training-fold risk counts N_k
Minimum Ordinal Proof Certificate (inference)
schematic selected witnesses M*[k]
Per boundary: selected indices • proof size m*[k]
dense / retained / complement: cD, cR, cC
sufficiency gap • complement drop
optional fixed-proof intervention Δ[i,k]
integrity hash + independent numerical replay
Ordinal Outputs (inference)
c[0..3] • P(Y>k) • P(Y=0..4)
upper median(q) → predicted grade ŷ
Fine-grid local witnesses: stride-8 centers with 95×95 receptive-field support — not pixel segmentation • Prediction has no global classifier bypass